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VIRILE AGITUR



Student Number

Knox Grammar School

2009

**Trial Higher School Certificate
Examination**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time - 3 hours
- Write using blue or black pen only
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Subject Teachers

Mr R. Maxwell
Mrs M. Wassef

This paper MUST NOT be removed from the examination room

Number of Students in Course: 40

**Number of Writing Booklets Per Student
(Eight Page) 8**

Total Marks – 120

- Attempt Questions 1 – 8
- Answer each question in a separate writing booklet
- All questions are of equal value

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Total marks – 120

Attempt Questions 1–8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks)

Marks

(a) Find $\int \frac{dx}{\sqrt{16x^2 - 1}}$

2

(b) Evaluate $\int_1^e x \ln x \, dx$

3

(c) (i) Find real numbers a and b such that

2

$$\frac{5x^2 + x + 8}{(x+1)(x^2+3)} \equiv \frac{a}{x+1} + \frac{bx-1}{x^2+3}$$

(ii) Hence find $\int \frac{5x^2 + x + 8}{(x+1)(x^2+3)} \, dx$

2

(d) Find $\int \tan^3 x \, dx$

2

(e) Using a suitable substitution, or otherwise, evaluate:

4

$$\int_0^2 \frac{x^2}{\sqrt{4-x^2}} \, dx$$

End of Question 1

Question 2 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Let $\alpha = 1 - \sqrt{3}i$.

(i) Find the exact value of $|\alpha|$ and $\arg \alpha$.

2

(ii) Hence express $(1 - \sqrt{3}i)^{10}$ in modulus-argument form.

1

(b) Express $\sqrt{7 - 24i}$ in the form $a + ib$, where a and b are real.

3

(c) Sketch the region in the complex plane where the two inequalities $0 \leq \arg(z) \leq \frac{3\pi}{4}$ and $|z - 2i| \geq |z|$ both hold.

3

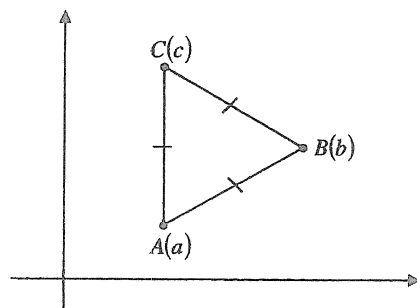
(d) Sketch the locus of z satisfying $|z - 3| + |z + 3| = 10$. Show any intercepts with the axes.

3

Question 2 continues on page 4

Question 2 (continued)

(e)



The points A , B and C on the Argand diagram represent the complex numbers a , b and c respectively. $\triangle ABC$ is equilateral.

Let $w = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$.

(i) Show that $\frac{a-b}{c-b} = w$.

1

(ii) By writing another similar expression for w , prove that

2

$$a^2 + b^2 + c^2 = ab + bc + ca$$

End of Question 2

Question 3 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) The equation $x^3 + 3x^2 - 5x - 2 = 0$ has roots α , β and γ .

2

Find a cubic equation with integer coefficients whose roots are $\frac{2}{\alpha}$, $\frac{2}{\beta}$ and $\frac{2}{\gamma}$.

(b) Consider the curve $x^2 + y^2 + xy = 3$.

(i) Show that $\frac{dy}{dx} = -\left(\frac{2x+y}{x+2y}\right)$.

1

(ii) Hence find the coordinates of any stationary points.

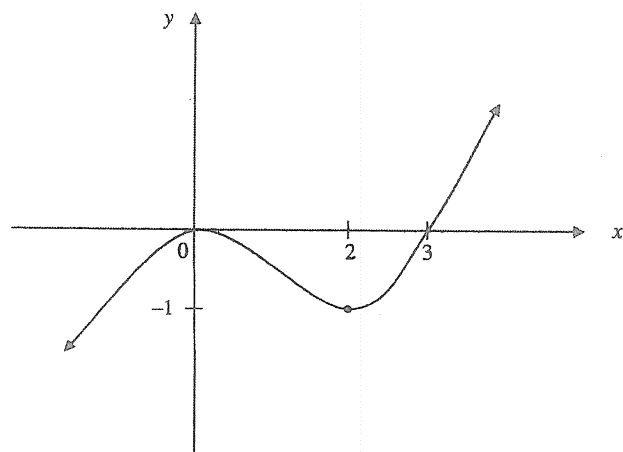
2

Question 3 continues on page 6

Question 3 (continued)

Marks

- (c) The diagram shows the graph of $y = f(x)$ where $f(x) = \frac{1}{4}x^2(x-3)$.



On the answer page provided, draw separate sketches of the graphs of the following:

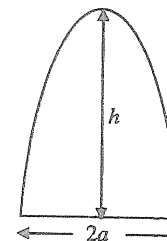
- | | | |
|-------|----------------------------|---|
| (i) | $y = \frac{1}{4}x^2 x-3 $ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 1 |
| (iii) | $y^2 = -f(x)$ | 2 |
| (iv) | $y = \tan^{-1}(f(x))$ | 2 |

Question 3 continues on page 7

Question 3 (continued)

Marks

- (d) (i)

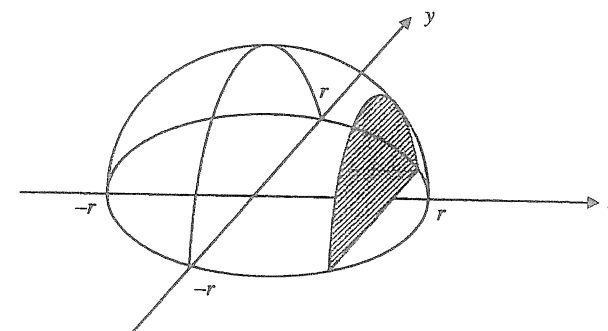


1

A parabolic segment has height h and width $2a$.

Use Simpson's Rule with three function values, to show that the exact area of this segment is $\frac{4ah}{3}$.

- (ii)



3

The base of a solid is the region in the xy plane enclosed by the circle $x^2 + y^2 = r^2$.

Each cross-section perpendicular to the x -axis is a parabolic segment with height one half its width.

Show that the volume of the solid is $\frac{16r^3}{9}$ units³.

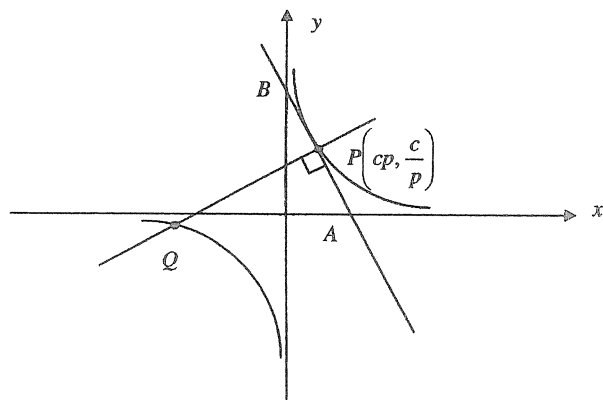
End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The point $P\left(cp, \frac{c}{p}\right)$ is a point on the hyperbola $xy = c^2$.

The tangent to the hyperbola at P intersects the x and y axes at A and B respectively and the normal to the hyperbola at P intersects the second branch at Q .



- (i) Show that the equation of the normal at P is $py - c = p^3(x - cp)$. 2

- (ii) Show that the x coordinates of P and Q satisfy the equation 2

$$x^2 - c\left(p - \frac{1}{p^3}\right)x - \frac{c^2}{p^2} = 0$$

and hence find the coordinates of Q .

- (iii) Given the distance $AB = 2c\sqrt{p^2 + \frac{1}{p^2}}$, show that the 2

$$\text{area of } \triangle ABQ = c^2\left(p^2 + \frac{1}{p^2}\right)^2.$$

- (iv) Find the minimum area of $\triangle ABQ$. 1

(You may use the inequality $\frac{a}{b} + \frac{b}{a} \geq 2$ for $a, b > 0$.)

Question 4 continues on page 9**Question 4 (Continued)**

Marks

- (b) An ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has foci S and S' where $b^2 = a^2(1 - e^2)$ where $0 < e < 1$ is its eccentricity.

- (i) Show that the equation of the normal to the ellipse at the point $P(x_1, y_1)$ is 2

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2.$$

- (ii) Show that N , the x -intercept of the normal has coordinates $N(e^2x_1, 0)$. 1

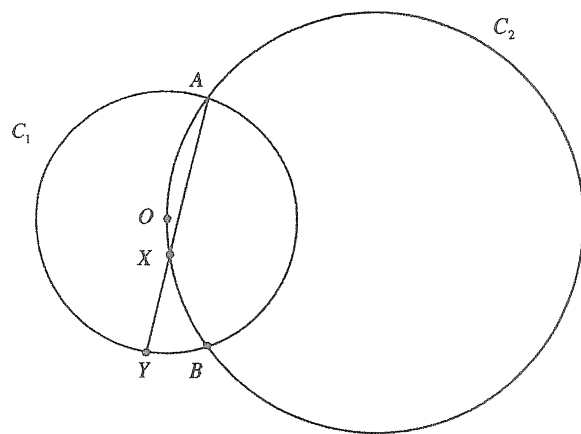
- (iii) Use the focus/directrix definition of a conic to prove that $\frac{SN}{S'N} = \frac{SP}{S'P}$ 2

Question 4 continues on page 10

Question 4 (continued)

Marks

(c)



Two circles C_1 and C_2 intersect at A and B . C_2 passes through O , the centre of C_1 . X lies on the arc AOB and AX intersects C_1 again at Y .

(i) State why $\angle AOB = 2 \times \angle AYB$.

1

(ii) Prove that $XY = XB$.

3

Question 5 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) (i) Show that if α is a double root of $f(x) = 0$ then $f(\alpha) = f'(\alpha) = 0$.

2

(ii) Find all roots of the equation $2x^3 - 5x^2 - 4x + 12 = 0$ given that two of the roots are equal.

3

(b) (i) By drawing a diagram, or otherwise, find the solutions of $z^5 = 1$.

2

(ii) Show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$.

2

(iii) Hence find the exact value of $\cos \frac{2\pi}{5}$.

2

(c) 11 persons gather to play basketball by forming 2 teams of 5 to play each other. The remaining person acts as a referee.

(i) In how many ways can the teams be formed?

2

(ii) If two particular persons are not to be in the same team, how many ways are there then to choose the teams?

2

Question 6 (15 marks) Use a SEPARATE writing booklet.

Marks

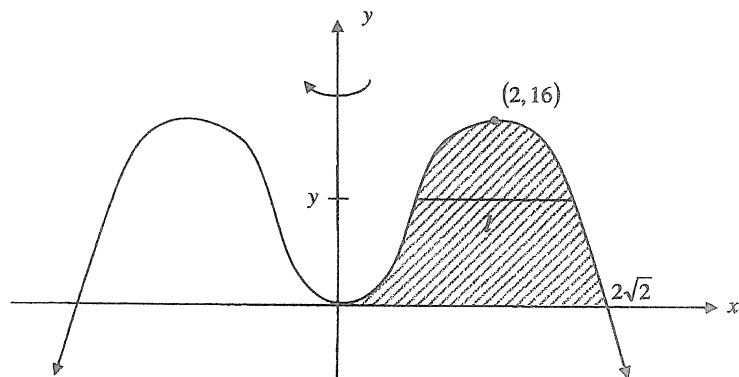
- (a) The sequence
- $\{a_n\}$
- is given by:

3

$$a_1 = 2, a_2 = \frac{3}{2} \text{ and } (n+1)a_{n+1} = a_{n-1} - (n-2)a_n \text{ for } n > 1.$$

Prove by induction that for $n \geq 1$, $a_n = \frac{n+1}{n!}$

- (b) The region bound by the curve $y = 8x^2 - x^4$ and the x axis in the first quadrant is rotated about the y axis to form a solid. When the region is rotated, the horizontal line segment l at height y sweeps out an annulus.



- (i) Show that the area of the annulus at height y is given by $2\pi\sqrt{16-y}$.

3

- (ii) Find the volume of the solid.

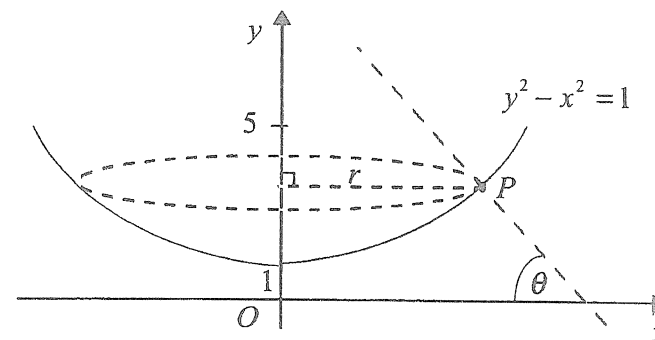
2

Question 6 continues on page 13

Question 6 (continued)

Marks

- (c)



A bowl is formed by rotating the hyperbola $y^2 - x^2 = 1$ for $1 \leq y \leq 5$ through 180° about the y -axis. Sometime later, a particle P of mass m moves around the inner surface of the bowl in a horizontal circle with constant angular velocity ω .

- (i) Show that if the radius of the circle in which P moves is r , then the normal to the surface at P makes an angle θ with the horizontal where $\tan \theta = \frac{\sqrt{1+r^2}}{r}$.
- (ii) Find expressions for the radius r of the circle of motion and the magnitude of the reaction force between the surface and the particle in terms of m, g and ω .
- (iii) Find the values for ω for which the described motion of P is possible.

3

2

3

End of Question 6

Question 7 (15 Marks) Use a SEPARATE writing booklet.

Marks

- (a) A particle of mass m is projected vertically upwards with an initial velocity of V in a medium where the resistance force R to the motion has a magnitude $R = mkv$ where v is the velocity of the particle after the initial projection.
- (i) Show that the maximum height h of the particle is given by 3
- $$h = \frac{g}{k^2} \left\{ \frac{k}{g} V - \ln \left(1 + \frac{k}{g} V \right) \right\}$$
- (ii) Find an expression for the time T of particle to reach its maximum height in terms of V , k and g . 2
- (iii) After reaching its maximum height the particle returns vertically downwards towards its projection point in the same medium. Show that the downward velocity is given by $v = \frac{g}{k} (1 - e^{-kt})$ where t is the time of the downward motion and give the terminal velocity of the particle. 3
- (iv) The speed of upward projection is double the terminal velocity and the particle's downward displacement y from its maximum height is given by the equation $-\frac{k^2}{g} y = \frac{k}{g} v + \ln \left(1 - \frac{k}{g} v \right)$. (Do NOT prove this result). 3
- Show that the velocity of the particle on return to its projection point is given by $\frac{kv}{g} + 2 - \ln 3 + \ln \left(1 - \frac{kv}{g} \right) = 0$. If a root to this equation for $\frac{kv}{g}$ lies near 0.81, use one application of "Newton's Method" to find a better approximation correct to two decimal places and deduce the percentage of the terminal velocity that the particle has acquired on return to its projection point. 3
- (v) Hence find the ratio of the time taken to reach maximum height to the time to fall from maximum height to the point of projection. 1
- (b) When an unbiased coin is tossed $2n$ times, the probability of observing k heads and $2n - k$ tails is given by $P_k = {}^{2n}C_k \left(\frac{1}{2} \right)^k \left(\frac{1}{2} \right)^{2n-k}$. 2
- (i) Show that the most likely outcome is $k = n$ 1
- (ii) Show that $P_n = \frac{(2n)!}{2^{2n} (n!)^2}$

End of Question 7

Question 8 (15 Marks) Use a SEPARATE writing booklet.

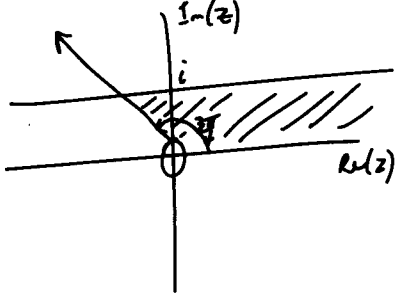
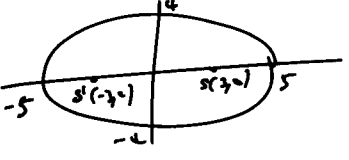
Marks

- (a) Given $I_n = \int_1^a (1 - \ln x)^n dx$ for $n = 0, 1, 2, \dots$
- (i) Show that $I_n = -1 + nI_{n-1}$ for $n = 1, 2, 3, \dots$ 2
- (ii) Hence evaluate $\int_1^e (1 - \ln x)^3 dx$ 2
- (iii) Show that $\frac{I_n}{n!} = e - \sum_{r=0}^n \frac{1}{r!}$ for $n = 1, 2, 3, \dots$ 2
- (iv) Show that $0 \leq I_n \leq e - 1$ 1
- (v) Deduce that $\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{r!} = e$ 1
- (b) $\phi(x)$ and $\psi(x)$ are continuous and bounded functions.
- (i) By considering $\int_0^a \{ \lambda \phi(x) + \psi(x) \}^2 dx$ for $a > 0$ as a quadratic function in λ , show that $\left\{ \int_0^a \phi(x) \psi(x) dx \right\}^2 \leq \int_0^a \{ \phi(x) \}^2 dx \times \int_0^a \{ \psi(x) \}^2 dx$. 4
- (ii) Hence show that $\left\{ \int_0^1 \phi(x) dx \right\}^2 \leq \int_0^1 \{ \phi(x) \}^2 dx$. 2
- (iii) Deduce that $\left\{ \int_0^1 \phi(x) dx \right\}^4 \leq \int_0^1 \{ \phi(x) \}^4 dx$. 1

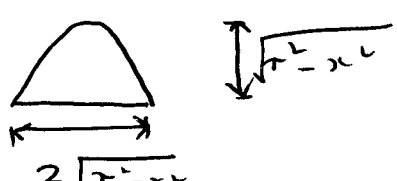
End of Paper

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| 1. WAM | 5. WAM |
| 2. MAX | 6. MAX |
| 3. HAR | 7. WAM |
| 4. DIN | 8. MAX |

2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 1</u> a) $\int \frac{dx}{\sqrt{16x^2-1}} = \frac{1}{4} \int \frac{dx}{\sqrt{x^2 - (\frac{1}{4})^2}}$ $= \frac{1}{4} \ln(x + \sqrt{x^2 - \frac{1}{4}}) + C$ $= \frac{1}{4} \ln(4x + \sqrt{16x^2-1}) + C$ b) $\int_1^e x \ln x$ $u = \ln x \quad v' = x$ $= \left[\frac{x^2 \ln x}{2} \right]_1^e - \int_1^e \frac{x}{2} dx \quad u' = \frac{1}{x}, v = \frac{x^2}{2}$ $= \frac{e^2}{2} - \left[\frac{x^2}{4} \right]_1^e \quad \checkmark$ $= \frac{e^2}{4} + \frac{1}{4} \quad \checkmark$ c) i) $a=3 \quad \checkmark \quad b=2 \quad \checkmark$ ii) $\int \frac{5x^2+x+8}{(x+1)(x+3)} dx = \int \frac{3}{x+1} + \frac{2x-1}{x^2+3} dx$ $= 3 \ln x+1 + \ln(x^2+3) - \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + C$ d) $\int \tan^3 x dx = \int \tan x (\sec^2 x - 1) dx$ $= \int \sec^2 x \tan x - \frac{\sin x}{\cos x} dx$ $= \frac{1}{2} \tan^2 x + \ln \cos x + C$ e) $\int_0^2 \frac{x^2}{\sqrt{4-x^2}} dx$ $\text{let } x = 2 \sin \theta$ $= \int_0^{\frac{\pi}{2}} \frac{4 \sin^2 \theta}{2 \cos \theta} \cdot 2 \cos \theta d\theta \quad dx = 2 \cos \theta d\theta$ $= \int_0^{\frac{\pi}{2}} 4 \sin^2 \theta d\theta = \int_0^{\frac{\pi}{2}} 2(1 - \cos 2\theta) d\theta$ $= \left[2\theta - \sin 2\theta \right]_0^{\frac{\pi}{2}} = \pi \quad \checkmark$	$\checkmark$ coeff of $\frac{1}{4}$ $\checkmark$ answer or equiv.	<u>Question 2</u> a) i) $z =2 \quad \arg(z) = -\frac{\pi}{3}$ ii) $(1-3i)^{10} = (2 \operatorname{cis}(-\frac{\pi}{3}))^{10}$ $= 2^{10} \operatorname{cis}(-\frac{10\pi}{3})$ $= 2^{10} \operatorname{cis}(\frac{2\pi}{3}) \quad \checkmark$ b) let $(a+bi)^2 = 7-24i$ $\therefore a^2 - b^2 + 2abi = 7-24i \quad \checkmark$ ie, $a^2 - b^2 = 7 \quad 2ab = -24$ $ab = -12$ by inspection $a = \pm 4 \quad a, b \in \mathbb{R}$ $b = \mp 3$ $\therefore \sqrt{7-24i} = 4-3i, -4+3i$ c)  $\checkmark 0 \leq \arg(z) \leq \frac{3\pi}{4}$ $\checkmark z-2i = z$ $\checkmark$ region deduced & origin included. d)  $\checkmark$ ellipse with foci $(\pm 3, 0)$ $\checkmark$ x-intercepts ± 5 $\checkmark$ y-intercepts ± 2 e) i) $\vec{BA} = (a-b), \vec{BC} = (c-b)$ Since $\vec{BA} = \vec{BC} \times \operatorname{cis}(\frac{\pi}{3}) \quad \checkmark$ $(a-b) = (c-b) \times \omega$ $\therefore \omega = \frac{a-b}{c-b}$ ii) Similarly $\frac{c-a}{b-a} = \omega \quad \checkmark$ $\therefore \frac{a-b}{c-b} = \frac{c-a}{b-a} \Rightarrow (a-b)(b-a) = (c-a)(c-b)$ $ab - a^2 - b^2 + ab = c^2 - cb - ac + ab$ $\therefore a^2 + b^2 + c^2 = ab + bc + ca$ $\checkmark$ progress to.	

2009 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 3</u> a) let $y = \frac{z}{x}$ $(\frac{z}{x})^3 + 3(\frac{z}{x})^2 - 5(\frac{z}{x}) - 2 = 0$ ✓ $\Rightarrow y^3 + 5y^2 - 6y + 4 = 0$ ✓ b) $x^2 + y^2 + xy = 3$ i) $2x + 2y \cdot \frac{dy}{dx} + (x \frac{dy}{dx} + y) = 0$ ✓ $(x + 2y) \frac{dy}{dx} = -2x - y$ $\frac{dy}{dx} = \frac{-(2x + y)}{x + 2y}$ ii) For stat. pts $\frac{dy}{dx} = 0$ $y = -2x$ $\therefore x^2 + (-2x)^2 + x(-2x) = 3$ $3x^2 = 3$ $x = \pm 1$ $y = \mp 2$ $\therefore$ stat pts $(1, -2)$ and $(-1, 2)$ ✓ c) see separate pages		d) i) $A = \frac{h}{3} [y_1 + 4y_2 + y_3]$ $= \frac{a}{3} [0 + 4h + 0]$ $= \frac{4ah}{3}$ ✓ ii) Cross-section:  Area = $\frac{4ah}{3} = \frac{4}{3} (r^2 - x^2)$ $V = 2 \int_0^r \frac{4}{3} (r^2 - x^2) dx$ ✓ $= \frac{8}{3} \left[r^2 x - \frac{x^3}{3} \right]_0^r$ ✓ $= \frac{16r^3}{9} \text{ u}^3$ ✓	

HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION
Mathematics Extension 2

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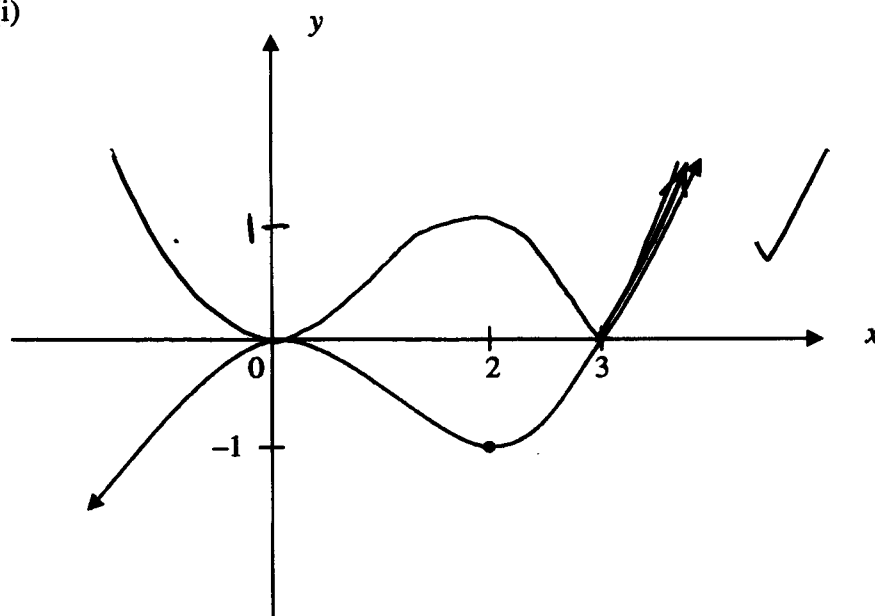
Centre Number

Questions 3 (c)

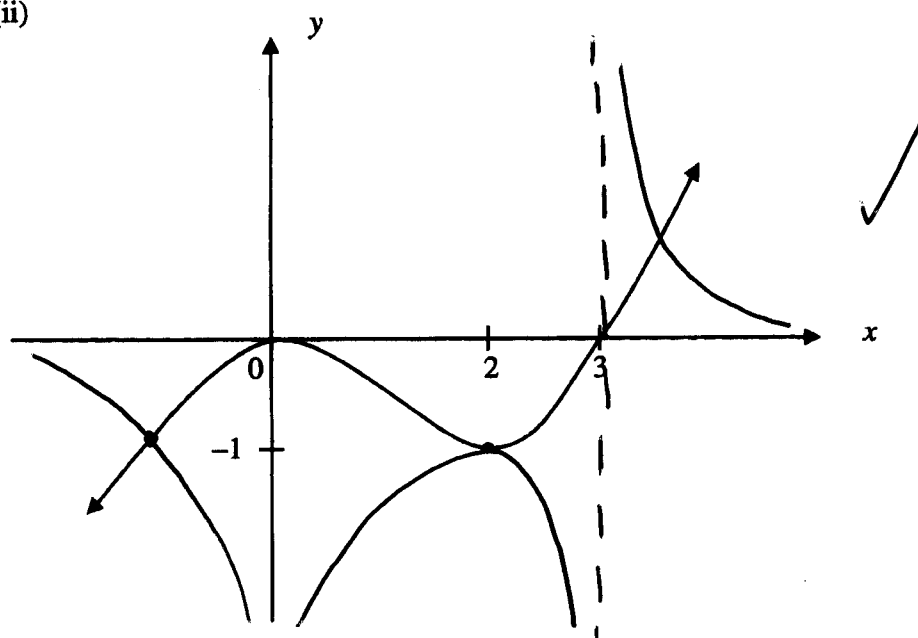
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Student Number

(i)



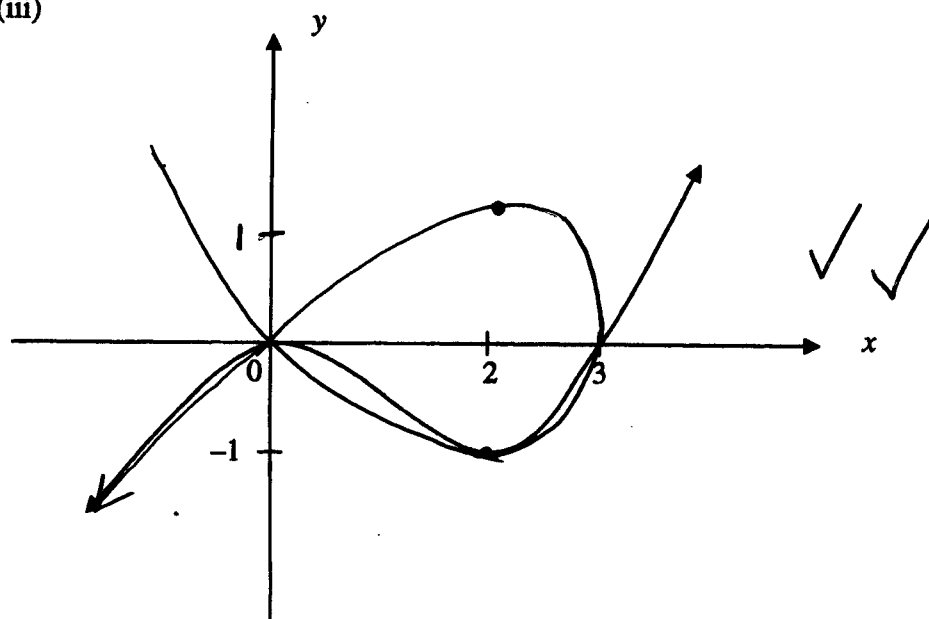
(ii)



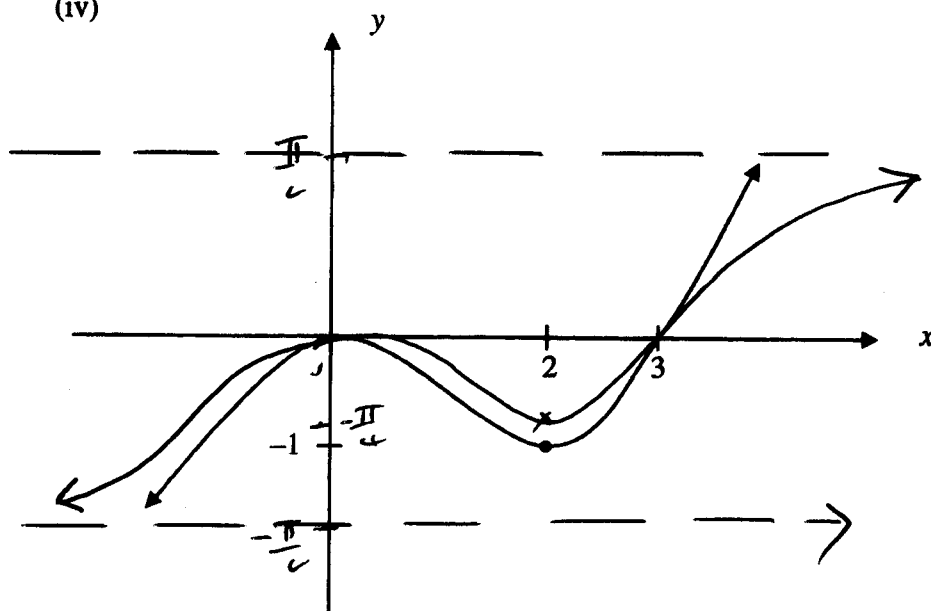
Question 3(c) continues on next page

Question 3 (continued)

(c) (iii)



(iv)



Question 4

a) i) $xy = c^2$
 $x \cdot \frac{dy}{dx} + y = 0$
 $\frac{dy}{dx} = -\frac{y}{x}$

at $P(cp, \frac{c}{p})$ $\frac{dy}{dx} = -\frac{\frac{c}{p}}{cp}$

$\therefore m_T = -\frac{1}{p^2}$

$m_T = p^2$ ✓

$\therefore y - \frac{c}{p} = p^2(x - cp)$

$py - c = p^3(x - cp)$ ✓

ii) Solving $y = \frac{c^2}{x}$ — ①

$py - c = p^3(x - cp)$ — ②

$p \cdot \frac{c^2}{x} - c = p^3(x - cp)$

$pc^2 - cx = p^3x^2 - cp^4x$

$\div p^3:$ $\frac{c^2}{p^2} - \frac{c}{p^2}x = x^2 - cp^2x$

$x^2 - c(p - \frac{1}{p^2})x - \frac{c^2}{p^2} = 0$

Let x coord of Q be α

$\therefore \alpha + cp = c(p - \frac{1}{p^2})$

$\therefore \alpha = -\frac{c}{p^3}$

sum of roots

$\therefore Q(-\frac{c}{p^3}, -cp^3)$ ✓

✓ - correct
progress
to
equation

$$\text{iii)} \quad AB = 2c \sqrt{\rho^2 + \frac{1}{\rho^2}} \quad (\text{given})$$

$$\therefore PQ = \sqrt{\left(c\rho + \frac{c}{\rho^3}\right)^2 + \left(\frac{c}{\rho} + c\rho^3\right)^2}$$

$$= c \sqrt{\rho^2 + \frac{2}{\rho^2} + \frac{1}{\rho^6} + \frac{1}{\rho^2} + 2\rho^2 + \rho^6}$$

$$= c \sqrt{\rho^6 + 3\rho^2 + \frac{3}{\rho^2} + \frac{1}{\rho^6}}$$

$$= c \sqrt{\left(\rho^2 + \frac{1}{\rho^2}\right)^3} \quad \checkmark$$

$$\text{Area } \triangle ABQ = \frac{1}{2} \times AB \times PQ$$

$$= \frac{1}{2} \times 2c \sqrt{\rho^2 + \frac{1}{\rho^2}} \times c \sqrt{\left(\rho^2 + \frac{1}{\rho^2}\right)^3}$$

$$= c^2 \left(\rho^2 + \frac{1}{\rho^2}\right)^2 \quad \checkmark$$

$$\text{iv)} \quad \text{if } \frac{a}{b} + \frac{b}{a} \geq 2$$

$$\rho^2 + \frac{1}{\rho^2} \geq 2$$

$$\therefore \text{Area } \triangle ABQ \geq c^2 \times 2^2$$

$$\geq 4c^2$$

$$\therefore \text{Min. Area is } 4c^2. \quad \checkmark$$

$$(b) \quad i) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\text{at } P(x_1, y_1) \quad \frac{dy}{dx} = -\frac{b^2 x_1}{a^2 y_1} \quad \checkmark$$

$$\therefore \text{gradient of normal} = +\frac{a^2 y_1}{b^2 x_1} \quad \checkmark$$

$$y - y_1 = \frac{a^2 y_1}{b^2 x_1} (x - x_1)$$

$$\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

$$ii) \quad \text{at } y = 0$$

$$a^2 x = x_1 (a^2 - b^2)$$

$$a^2 x = x_1 (a^2 - a^2(1 - e^2))$$

$$a^2 x = x_1 a^2 e^2$$

$$\therefore x = e^2 x_1 \quad \checkmark$$

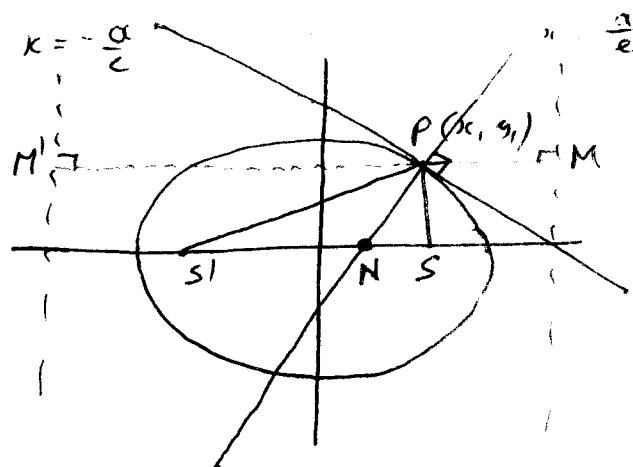
$$\therefore N(e^2 x_1, 0)$$

$$iii) \quad R.H.S = \frac{PS}{PS'}$$

$$= \frac{e PM}{e PM'}$$

$$= \frac{\frac{a}{e} - x_1}{\frac{a}{e} + x_1}$$

$$= \frac{a - ex_1}{a + ex_1} \quad \checkmark$$



$$L.H.S = \frac{SN}{S'N}$$

$$= \frac{ae - e^2 x_1}{ae - e^2 x_1} = \frac{e(a - ex_1)}{e(a - ex_1)}$$

$$\checkmark \therefore \frac{SN}{S'N} = \frac{SP}{S'P}$$

4(c) i) If $\angle XYB = \alpha$
 $\angle AOB = 2\alpha$ (L at centre is double L ^{at circumference} standing on same arc). ✓

$\angle AXB = \angle AOB = 2\alpha$ (L's in same segment are equal). ✓

$$\begin{aligned}\therefore \angle XBY &= \angle AXB - \angle XYB \\ &= 2\alpha - \alpha \\ &= \alpha\end{aligned}$$

(Ext L. of Δ equals sum of 2 opp. int. L's) ✓

$\therefore XY = XB$ (sides opposite equal L's in a Δ are equal)

Question 5

(a) (i) If α is a double root of $f(x) = 0$, then $f(x)$ can be written:

$$f(x) = (x - \alpha)^2 \cdot Q(x)$$

$$f'(x) = 2(x - \alpha) \cdot Q(x) + (x - \alpha)^2 \cdot Q'(x)$$

$$= (x - \alpha) [2Q(x) + (x - \alpha) \cdot Q'(x)]$$

$$= (x - \alpha) \cdot \Phi(x)$$

$$f(\alpha) = (\alpha - \alpha)^2 \cdot Q(\alpha) \quad f'(\alpha) = (\alpha - \alpha) \cdot \Phi(\alpha)$$

$$= 0 \cdot Q(\alpha)$$

$$= 0 \cdot \Phi(\alpha)$$

$$= 0$$

$$\therefore f(\alpha) = f'(\alpha) = 0$$

(ii) let $f(x) = 2x^3 - 5x^2 - 4x + 12$

$$\text{so } f'(x) = 6x^2 - 10x - 4 = 0$$

$$2x^2 - 5x - 2 = 0$$

$$(3x + 1)(x - 2) = 0$$

$$x = -\frac{1}{3}, 2$$

$$\text{and } f(2) = 0$$

$\therefore x = 2$ is the double root

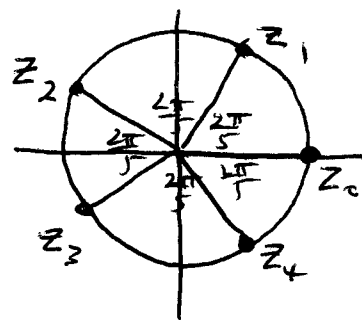
$$\text{so } 2 + 2 + \beta = \frac{5}{2}$$

$$\beta = -\frac{3}{2}$$

$\therefore$ Roots are $2, 2, -\frac{3}{2}$

5b) i)
$$\left. \begin{aligned} z_0 &= 1 \\ z_1 &= \text{cis } \frac{2\pi}{5} \\ z_2 &= \text{cis } \frac{4\pi}{5} \\ z_3 &= \text{cis } -\frac{4\pi}{5} \\ z_4 &= \text{cis } -\frac{2\pi}{5} \end{aligned} \right\}$$

✓ All roots



✓ diagram or otherwise

ii)

Sum of roots $= -\frac{b}{a}$

$\therefore z_0 + z_1 + z_2 + z_3 + z_4 = -\frac{0}{1}$

✓ $z^5 - 1 = 0$

Note $\bar{z}_1 = z_4 \quad \bar{z}_2 = z_3$

$1 + z_1 + \bar{z}_1 + z_2 + \bar{z}_2 = 0$

Note $z + \bar{z} = 2\text{Re}(z)$

$\therefore 1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0$

$\therefore \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2}$

iii)

$\cos\frac{2\pi}{5} + 2\cos^2\frac{2\pi}{5} - 1 = -\frac{1}{2}$

$4\cos^2\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} + 1 = 0$

$\therefore \cos\frac{2\pi}{5} = \frac{-1 \pm \sqrt{5}}{4}$ but $\cos\frac{2\pi}{5} > 0$

$\therefore \cos\frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}$

C i) $\frac{{}^{11}C_5 \times {}^6C_5 \times {}^1C_1}{2} = 1386$

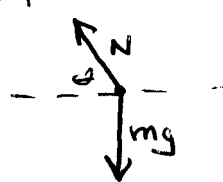
ii) ${}^9C_4 \times {}^5C_4 \times {}^1C_1 + \frac{{}^2C_1 \times {}^{10}C_5 \times {}^5C_5}{2}$
 $= 630 + 252 = 882$

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 6</u> a) When $n=1$, $\frac{1+1}{1!} = 2 = a_1$ When $n=2$, $\frac{2+1}{2!} = \frac{3}{2} = a_2$ $\therefore$ statement True for $n=1, 2$ Assume the statement true for $n=k-1$ and $n=k$ i.e. $a_{k-1} = \frac{k}{(k-1)!}$ $a_k = \frac{k+1}{k!}$ Let $n=k+1$ L.H.S. = a_n $= a_{k+1}$ $= \frac{a_{k-1} - (k-2)a_k}{(k+1)}$ $= \frac{\frac{k}{(k-1)!} - (k-2)\frac{k+1}{k!}}{(k+1)}$ $= \frac{k^2 - (k^2 - k - 2)}{k!(k+1)}$ $= \frac{k+2}{(k+1)!}$ $= \frac{n+1}{n!}$ where $n=k+1$	✓ ✓ ✓	✓ $(n+1)a_{n+1} = a_{n-1} - (n-2)a_n$ If the statement true for $n=k$, then proved true for $n=k+1$. Since true for $n=1, 2$, $\therefore$ true for $n=3, 4, 5, \dots$ $\therefore$ true for all $n \geq 1$.	

- | | |
|--------|--------|
| 1. WAM | 5. WAM |
| 2. MAX | 6. WAM |
| 3. HAR | 7. MAX |
| 4. DIN | 8. MAX |

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
6 b) i) $y = 8x^2 - x^4$ $x^4 - 8x^2 = -y$ $(x^2 - 4)^2 = 16 - y$ $x^2 = \pm \sqrt{\pm \sqrt{16 - y} + 4}$ let x_1 and x_2 be endpoints of l with $0 \leq x_1 \leq x_2$ $\therefore x_2 = +\sqrt{+\sqrt{16 - y} + 4}$ $x_1 = -\sqrt{-\sqrt{16 - y} + 4}$ $\therefore \text{Area} = \pi (x_2^2 - x_1^2)$ $= \pi (\sqrt{16 - y} + 4 - (-\sqrt{16 - y} + 4))$ $= 2\pi \sqrt{16 - y}$ ii) $V = \int_0^{16} 2\pi \sqrt{16 - y} \, dy$ $= \left[-\frac{4\pi}{3} (16 - y)^{\frac{3}{2}} \right]_0^{16}$ $= \frac{4^4 \pi}{3} \quad \checkmark$	$\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$	c) i) $y^2 - x^2 = 1$ $\therefore 2y \frac{dy}{dx} - 2x = 0$ $\therefore \frac{dy}{dx} = \frac{x}{y}$ at $P(r, \sqrt{1 + r^2})$ $\therefore m_{\text{Normal}} = -\frac{\sqrt{1 + r^2}}{r}$ $\therefore \tan \theta = \frac{\sqrt{1 + r^2}}{r}$ ii) Forces on P  Resolving Vertically and Horizontally $N \sin \theta = mg \quad \text{--- (1)}$ $N \cos \theta = m r \omega^2 \quad \text{--- (2)}$ $\text{(1)} \div \text{(2)} \quad \tan \theta = \frac{g}{r \omega^2}$ $\therefore \frac{\sqrt{1 + r^2}}{r} = \frac{g}{r \omega^2}$ $\therefore r = \frac{\sqrt{g^2 - \omega^4}}{\omega^2} \quad \checkmark$ From (2): $N^2 = m^2 r^2 \omega^4 \sec^2 \theta$ $= m^2 \omega^4 (r^2 + r^2 \tan^2 \theta)$ $= m^2 \omega^4 \left(\frac{g^2}{\omega^4} - 1 + \frac{g^2}{\omega^4} \right)$ $= m^2 (2g^2 - \omega^4)$ $\therefore N = \sqrt{2g^2 - \omega^4} \quad \checkmark$ iii) $\omega^4 \leq g^2 \quad \therefore \omega \leq \sqrt{g} \quad \checkmark$	$\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$

$$y \leq 5 \quad \therefore \sqrt{1 - r^2} \leq 5 \quad \therefore \frac{g}{\omega^2} \leq 5 \quad \checkmark$$

$$\therefore \sqrt{\frac{g}{5}} \leq \omega \leq \sqrt{g} \quad \checkmark$$

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 7</u> a) i) For upward motion $x \uparrow \quad mg \downarrow \quad R \downarrow$ $F = m\ddot{x} = -mg - Rv$ $\therefore \ddot{x} = -(g + kv)$ $v \frac{dv}{dx} = -(g + kv)$ $\frac{dx}{dv} = \frac{-v}{g + kv}$ ✓ $k \frac{dx}{dv} = -1 + \frac{g}{g + kv}$ $\therefore \frac{k^2}{g} \frac{dx}{dv} = -\frac{k}{g} + \frac{k}{g + kv}$ $\therefore \frac{k^2}{g} x = -\frac{k}{g} v + \ln(g + kv) + C$ at $x = 0, v = V$ $\therefore C = \frac{k}{g} V - \ln(g + kV)$ $\therefore \frac{k^2}{g} x = \frac{k}{g} (V - v) - \ln\left(\frac{g + kV}{g + kv}\right)$ or equivalent ✓ at $x = h, v = 0$ $\therefore h = \frac{g}{k^2} \left(\frac{k}{g} V - \ln\left(1 + \frac{k}{g} V\right) \right)$ ✓		ii) $\ddot{x} = -(g + kv)$ $\frac{dv}{dt} = -(g + kv)$ $\therefore \frac{dt}{dv} = -\frac{1}{g + kv}$ ✓ $t = -\frac{1}{k} \ln(g + kv) + C$ at $t = 0, v = V$ $\therefore t = \frac{1}{k} \ln\left(\frac{g + kV}{g + kv}\right)$ or equivalent at $v = 0, t = T$ $\therefore T = \frac{1}{k} \ln\left(1 + \frac{k}{g} V\right)$ or equivalent ✓ iii) Downward motion. $\downarrow mg \quad \uparrow R$ $\therefore F = m\ddot{x} = mg - mkv$ $\frac{dv}{dt} = g - kv$ $\frac{dt}{dv} = \frac{1}{g - kv}$ $t = -\frac{1}{k} \ln(g - kv) + C$ at $t = 0, v = 0 \therefore C = \frac{1}{k} \ln g$ ✓	

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
part iii) continued $\therefore t = \frac{1}{k} \ln\left(\frac{g}{g-kv}\right)$ or $-kt = \ln\left(\frac{g-kv}{g}\right)$ $\frac{g-kv}{g} = e^{-kt}$ $1 - \frac{k}{g}v = e^{-kt}$ $\therefore v = \frac{g}{k}(1 - e^{-kt})$ at $t \rightarrow \infty, e^{-kt} \rightarrow 0$ $v \rightarrow \frac{g}{k} \text{ (Terminal } v) \checkmark$ iv) If $V = \frac{2g}{k}$ $h = \frac{g}{k^2}(2 - \ln 3) \text{ from part i)}$ and sub in $-\frac{k}{g}y = \frac{k}{g}vt + \ln\left(1 - \frac{k}{g}v\right)$ $\therefore \frac{k}{g} \cdot \frac{g}{k^2}(2 - \ln 3) = \frac{k}{g}v + \ln\left(1 - \frac{k}{g}v\right)$ $\therefore \frac{kv}{g} + 2 - \ln 3 + \ln\left(1 - \frac{kv}{g}\right) = 0 \checkmark$		let $u = \frac{kv}{g}$ $\therefore f(u) = u + 2 - \ln 3 + \ln(1-u)$ $f'(u) = 1 - \frac{1}{1-u}$ at $u_1 = 0.81$ $u_2 = 0.81 - \frac{0.81 + 2 - \ln 3 + \ln(0.19)}{1 - \frac{1}{0.19}}$ $\doteq 0.82 \text{ (2 d.p.)} \checkmark$ $\therefore \frac{kv}{g} \doteq 0.82$ and $v \doteq 0.82 \frac{g}{k}$ or 82% of terminal velocity on return $\checkmark$ v) from part iii) $kt = -\ln\left(1 - \frac{kv}{g}\right)$ at $\frac{kv}{g} = 0.82, kT = -\ln(1 - 0.82)$ $= -\ln(0.18)$ from part ii) $V = \frac{2g}{k} \Rightarrow kT = \ln 3$ $\therefore \frac{-\ln(0.18)}{\ln 3} \doteq 0.64 \checkmark$	

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
7 (b) i) $\frac{P_{k+1}}{P_k} = \frac{\frac{(2n)!}{(k+1)!(2n-k-1)!} \left(\frac{1}{2}\right)^{k+1}}{\frac{(2n)!}{k!(2n-k)!} \left(\frac{1}{2}\right)^k} \left(\frac{1}{2}\right)^{2n-k-1}$ $= \frac{k!(2n-k)!}{(k+1)!(2n-k-1)!} \frac{\left(\frac{1}{2}\right)^{2n-k}}{\left(\frac{1}{2}\right)^{2n-k-1}}$ $= \frac{2n-k}{k+1} \checkmark$ Solve $P_{k+1} > P_k$ for greatest value of P_k ie, $2n-k > k+1$ $2k < 2n-1$ $k < n - \frac{1}{2}$ $\therefore P_{k+1} > P_k$ for $k=0, 1, 2, \dots, n-1$ and $P_{k+1} < P_k$ for $k=n+1, n+2, \dots, 2n$ $\therefore k=n$ gives greatest value for P_k and hence most likely.		ii) $P_n = {}^{2n}C_n \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^{2n-n}$ $= \frac{2n!}{n!n!} \cdot \left(\frac{1}{2}\right)^n \cdot \left(\frac{1}{2}\right)^n$ $= \frac{2n!}{(n!)^2} \checkmark$	

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 8</u> (a)i) Let $u = (1 - \ln x)^n$ $\therefore u' = \frac{n(1 - \ln x)^{n-1}}{-x}$ $\therefore I_n = \left[x(1 - \ln x)^n \right]_1^e + n \int_1^e (1 - \ln x)^{n-1} dx$ ✓ $\therefore I_n = -1 + n I_{n-1}$ ii) $I_0 = \int_1^e 1 dx$ $= e - 1$ ✓ $\therefore I_3 = -1 + 3I_2$ $= -1 + 3(-1 + 2I_1)$ $= -4 + 6(-1 + I_0)$ $= -16 + 6e$ ✓ iii) $\frac{I_n}{n!} = \frac{-1}{n!} + \frac{n I_{n-1}}{n!}$ $= \frac{-1}{n!} + \frac{I_{n-1}}{(n-1)!}$ ✓ $\sum_{r=1}^n \frac{I_r}{r!} = -\sum_{r=1}^n \frac{1}{r!} + \sum_{r=1}^n \frac{I_{r-1}}{(r-1)!}$ $= -\sum_{r=1}^n \frac{1}{r!} + \sum_{r=0}^{n-1} \frac{I_r}{r!}$	$v' = 1$ $v = x$	✓ $\therefore \frac{I_n}{n!} = -\sum_{r=1}^n \frac{1}{r!} + \frac{I_0}{0!}$ $= -\sum_{r=1}^n \frac{1}{r!} + \frac{-1}{0!} + \frac{e!}{0!}$ $= e - \sum_{r=1}^n \frac{1}{r!}$ ✓ reasoning to answer iv) $1 \leq x \leq e$ $\therefore 0 \leq \ln x \leq 1$ $0 \leq (1 - \ln x)^n \leq 1$ so $0 \leq \int_1^e (1 - \ln x)^n dx \leq \int_1^e 1 dx$ $0 \leq I_n \leq e - 1$ v) $0 \leq \frac{I_n}{n!} \leq \frac{e-1}{n!}$ as $n \rightarrow \infty, \frac{e-1}{n!} = 0$ $\therefore \lim_{n \rightarrow \infty} \frac{I_n}{n!} = 0$ ✓ and the deduce.	

so $\lim_{n \rightarrow \infty} \left(e - \sum_{r=0}^n \frac{1}{r!} \right) = \lim_{n \rightarrow \infty} \frac{I_n}{n!} = 0$
 $\therefore e - \lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{1}{r!} \right) = 0$
 $\lim_{n \rightarrow \infty} \sum_{r=0}^n \left(\frac{1}{r!} \right) = e$

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Q8 (b) i) Since $a > 0$, $\int_0^a \{\lambda \phi(x) + \psi(x)\}^2 dx \geq 0$ for all real λ ✓ $\therefore \int_0^a \{\lambda \phi(x) + \psi(x)\}^2 dx = \lambda^2 \int_0^a \{\phi(x)\}^2 dx + 2\lambda \int_0^a \phi(x)\psi(x) dx + \int_0^a \{\psi(x)\}^2 dx$ ✓ Considered as a quadratic in λ, $\Delta \leq 0$. $\therefore 4 \left\{ \int_0^a \phi(x)\psi(x) dx \right\}^2 - 4 \int_0^a \{\phi(x)\}^2 dx \cdot \int_0^a \{\psi(x)\}^2 dx \leq 0$ ✓ $\therefore \left\{ \int_0^a \phi(x)\psi(x) dx \right\}^2 \leq \int_0^a \{\phi(x)\}^2 dx \cdot \int_0^a \{\psi(x)\}^2 dx$ ✓			
ii) if $a = 1$ and $\psi(x) = 1$ ✓ $\int_0^1 1^2 dx = 1$ ✓ $\therefore \left\{ \int_0^1 \phi(x) dx \right\}^2 \leq \int_0^1 \{\phi(x)\}^2 dx$ ✓			
iii) $\left\{ \int_0^1 \phi(x) dx \right\}^4 \leq \left\{ \int_0^1 \{\phi(x)\}^2 dx \right\}^2 \leq \int_0^1 \left\{ \{\phi(x)\}^2 \right\}^2 dx$ ✓ $\therefore \left\{ \int_0^1 f(x) dx \right\}^4 \leq \int_0^1 \{f(x)\}^4 dx$.			